PORTFOLIO

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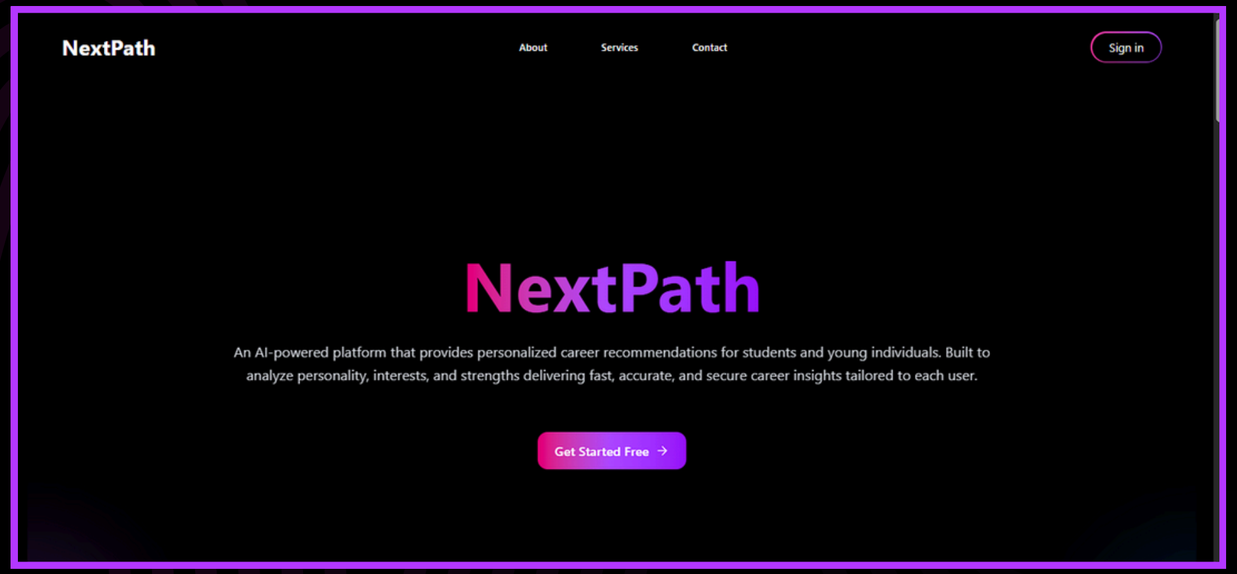
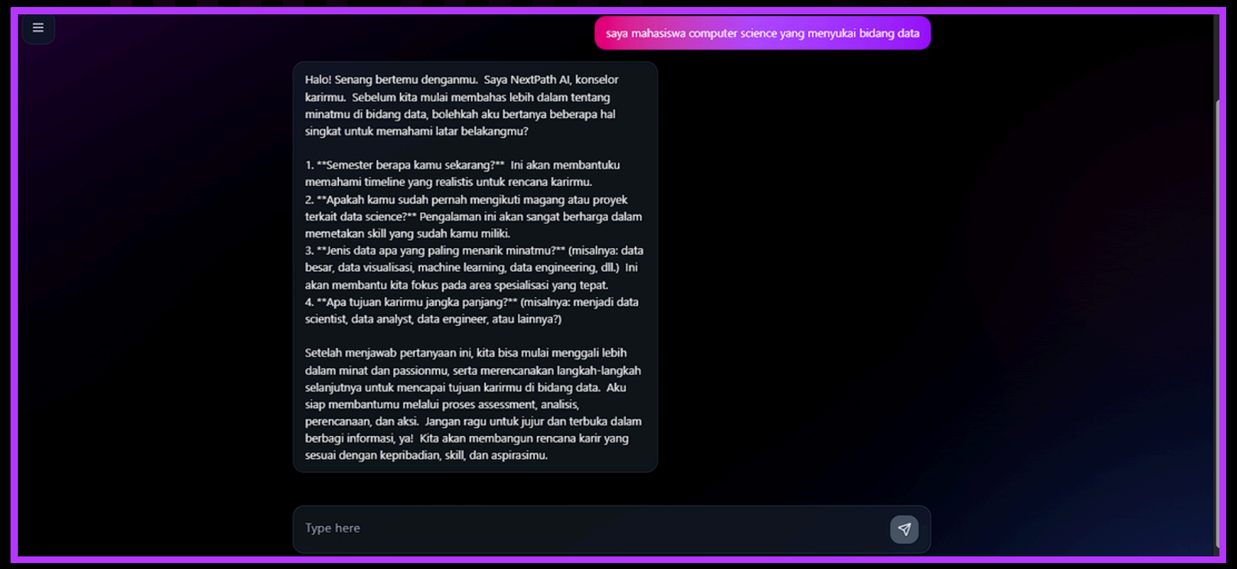
Intelligence System (AI) Streaming Program

PROJECT 1: NextPath

This is a **group project** for Informatics Festival 2025 at Universitas PGRI Semarang. This is a **web-based AI chatbot** that focuses on **career consultation**. It can give recommendation, pros and cons, detailed information, based on the user's conditio. This project is developed by a team consisting of 4 people. My part is developing the base LLM model for the website. I used Gemini open-source model and prompt-tuned it to be more focused on career solutions.
Website can be accessed here: https://nextpath.my.id/



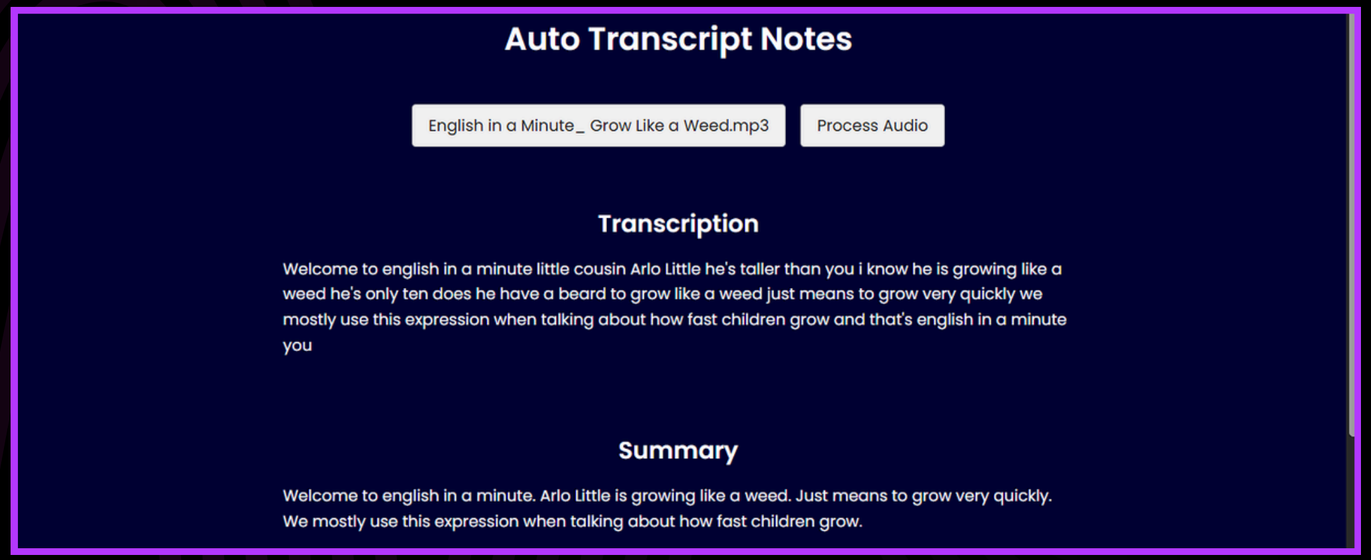


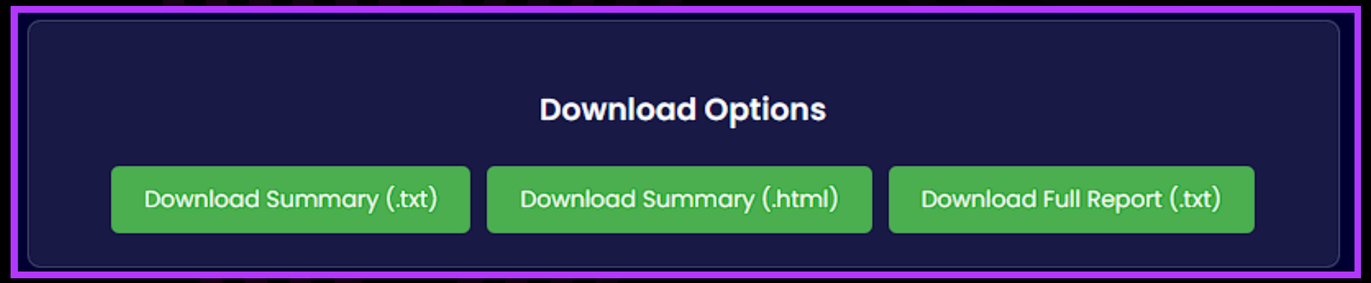
PROJECT 2: Lecture Transcription and Summarizer

This is a **group project** for Speech Recognition and Natural Language Processing class. This is an **Automatic Speech Recognition** model can transcript a lecture audio and summarize it into a text-based file (currently only .txt and .html). There's two option of outputs that the user can choose:

1. Summarization only
2. Original Transcription and Summarization

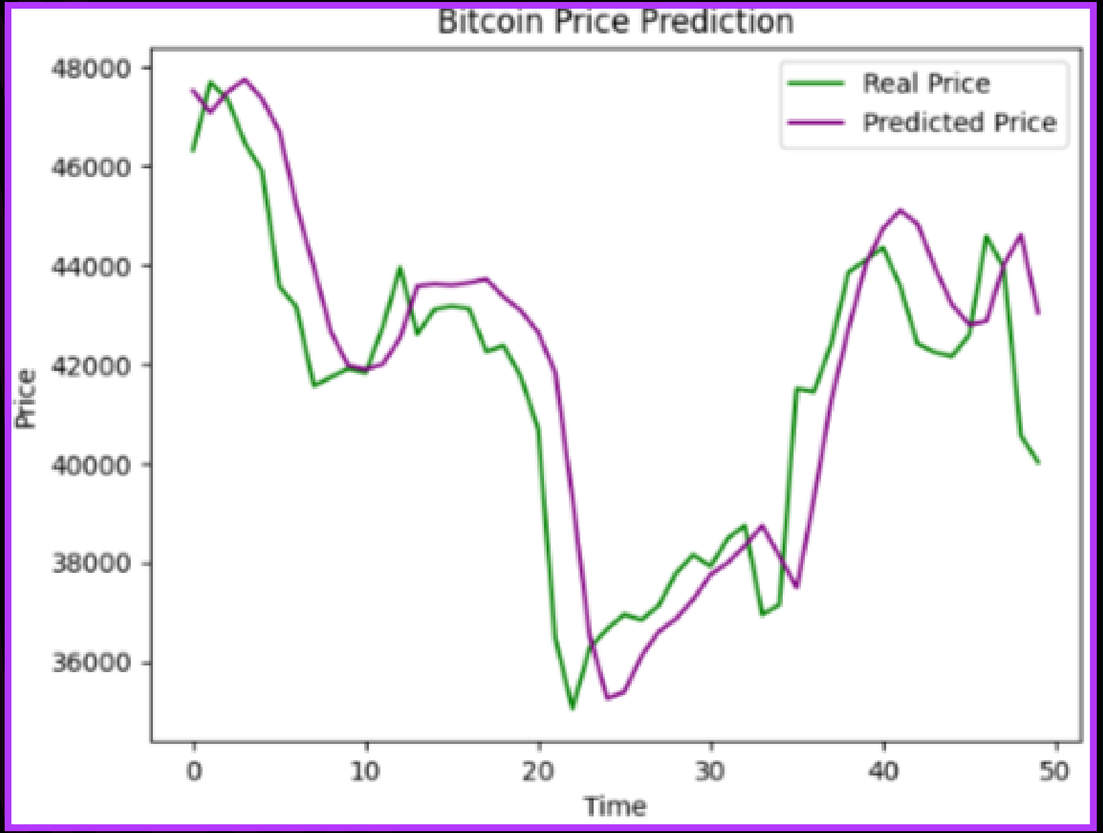
My part of this project is making the base model. I fine-tuned the Whisper (Base) Model with a 20-hour audio dataset. Then, I made the UI using React (HTML, CSS, JS).





PROJECT 3: Bitcoin Price Prediction (RNN & LSTM)

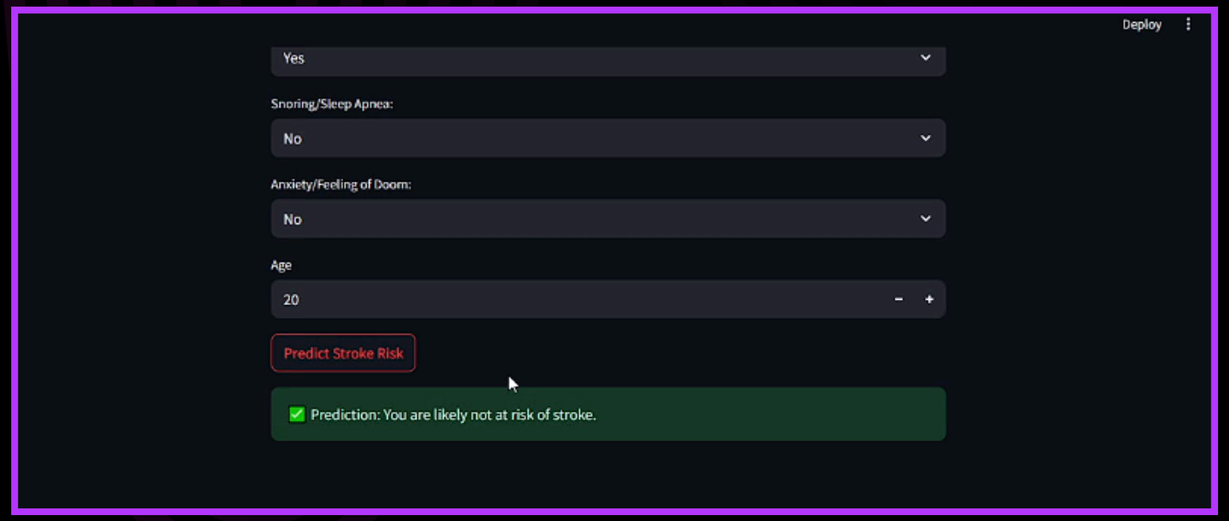
This is a **personal project** that I made to learn about Deep Learning. I used **Recurrent Neural Network (RNN) + LSTM** architecture to make a model for **predicting Bitcoin (and possibly) any other assets price**. Using the historical data of its previous 90 days 'open' price, the model can predict the price of the next 50 days. This project is still on-going. Further development is to make the model get more input features and make the prediction's timestep longer so that the prediction is more reliable. Also, the dataset I used is limited up to 20 February 2022, so it's not an up-to-date dataset.



Model Testing Result

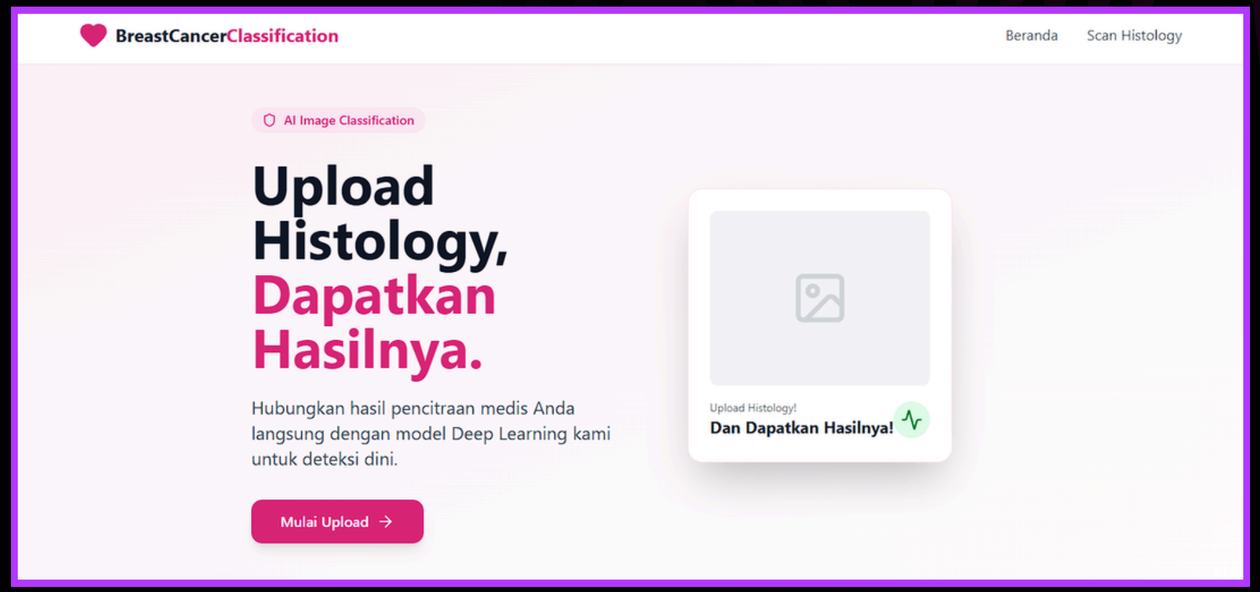
PROJECT 4: Stroke Risk Prediction

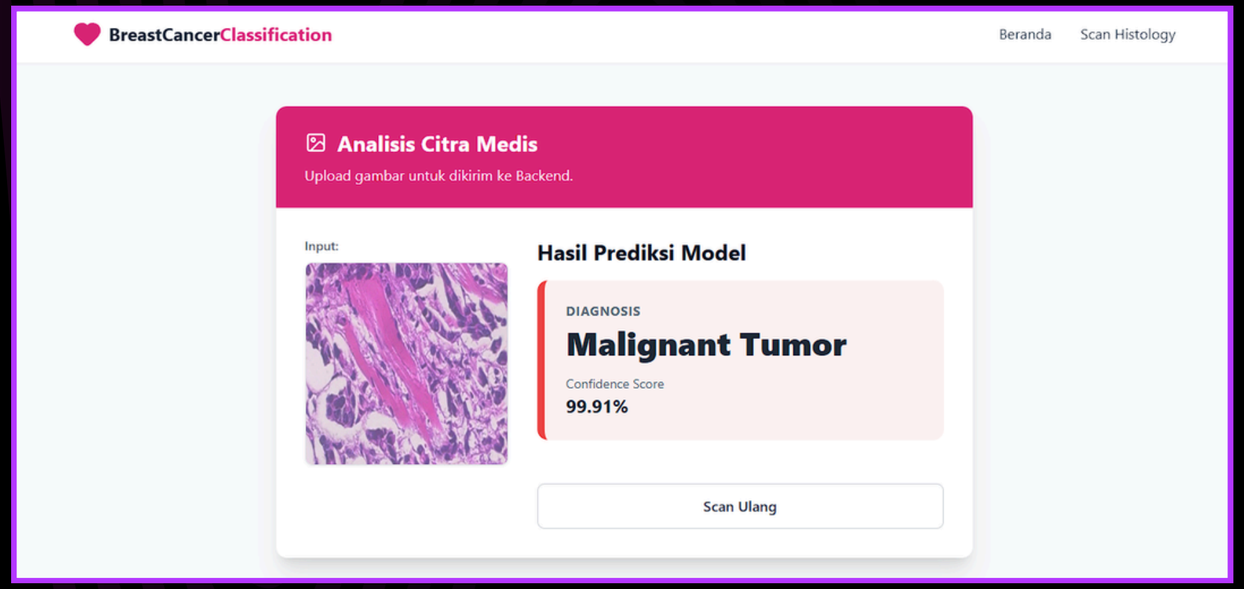
This is a group **project** for Research Methodology and Machine Learning Class, in a team consisting of two people. This model can predict whether someone is at risk of stroke or not, based on the symptoms they have. My part of this project is finding the dataset, developed the model and do the comparative analysis. I used 7 different models which are ANN, KNN, Naive Bayes Classifer, Random Forest, XGBoost, LightGBM, and CatBoost. The results show that CatBoost got the best performance of 100% accuracy. I did another test with an augmented dataset and unseen data to avoid overfitting, and turns out the result is still as good, 100% accuracy with zero False-Positives and False-Negatives. So, we decided to continue with CatBoost as the base model for the website.



PROJECT 5: Breast Cancer Image Classification

This is a **group project** for Deep Learning subject. My role in this project is Data Science and Deep Learning developer. I developed the model using Transfer Learning method with MobileNetV2. The evaluation shows 99% accuracy with very few False Negative on the confusion matrix, which is very important for medical classification cases.





THERE'S MORE!

But unfortunately, I can't show them all here.
Feel free to check my LinkedIn though.

You can access it here